Inorganic Cumulative Exam

September 14, 2012

Announced Topic: Applications of spectroscopy to structure determination in inorganic chemistry

1. Several varieties of hydrogenase contain irons centers coordinated to CO and/or CN^-. X-ray diffraction cannot reliably distinguish between the two. FTIR is the best choice for this determination. If an investigator proposed that the metal was coordinated to one CO and one CN in a cis geometry what tell-tale features of the FTIR would the investigator look for in order to prove his proposal. Be as specific as possible with regard to vibrational frequencies numbers of bands and band intensities if these are involved in your answer.

2. Carbon monoxide is a prominent ligand in organometallic chemistry. If an organometallic complex was prepared that had two CO's coordinated to one metal could FTIR be used to distinguish between cis and trans orientations of the CO's? If yes how.

3. In some organometallic compounds the coordinated CO may be a bridging ligand. How can FTIR be used to show the existence of this coordination geometry?

4. Numerous studies have been published relating the CO stretching frequency to the presence of other ligands coordinated to the metal center and how these ligands compete with the CO pi-backbonding. Predict what would happen to the CO stretching frequency if a strong pi backbonding ligand replaces a weak backbonding ligand in the complex. Explain the rationale for your prediction. This information can be used to confirm the presence of the new ligand.

5. FTIR spectra can be very complex and band assignments are not always reliable. How would you prove (or strongly support) that a particular band in an IR spectrum is due to a specific vibrational assignment?